# TRANSFORMASI LAPLACE DARI TURUNAN DAN INTEGRAL

No:
Date:

Latihan 4.3

Tentukan invers transformasi Laplace dengan menggunakan metode pecahan parsial untuk fungsi-fungsi berikut:

1. $F(s) = \frac{1}{s(s+2)}$

$$\frac{1}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2}$$

$$\frac{1}{s(s+2)} = \frac{A(s+2) + Bs}{s(s+2)}$$

$$A(s+2) + Bs = 1$$

$$As + 2A + Bs = 1$$

$$(A+B)s + 2A = 1$$

$$\begin{cases} A + B = 0 \rightarrow B = -A \rightarrow B = -\frac{1}{2} \\ 2A = 1 \rightarrow A = \frac{1}{2} \end{cases}$$

$$f(t) = L^{-1}\left\{\frac{\frac{1}{2}}{s} - \frac{\frac{1}{2}}{s+2}\right\}$$

$$= \frac{1}{2}L^{-1}\left[\frac{1}{s}\right] - \frac{1}{2}L^{-1}\left[\frac{1}{s-2}\right]$$

$$= \frac{1}{2} - \frac{1}{2}e^{2t}$$

2. $F(s) = \frac{1}{(s-1)(s-2)}$

$$\frac{1}{(s-1)(s-2)} = \frac{A}{s-1} + \frac{B}{s-2}$$

$$\frac{1}{(s-1)(s-2)} = \frac{A(s-2) + B(s-1)}{(s-1)(s-2)}$$

$$A(s-2) + B(s-1) = 1$$

$$As - 2A + Bs - B = 1$$

$$(A+B)s - 2A - B = 1$$

$$\begin{cases} A + B = 0 \rightarrow A = -B \\ -2A - B = 1 \rightarrow 2B - B = 1 \rightarrow B = 1 \text{ dan } A = -1 \end{cases}$$

$$f(t) = L^{-1}\left\{-\frac{1}{s-1} + \frac{1}{s-2}\right\}$$

$$= L^{-1}\left[-\frac{1}{s-1}\right] + L^{-1}\left[\frac{1}{s-2}\right]$$

$$= -e^{t} + e^{2t}$$

3. $F(s) = \frac{2s+1}{(s+1)(s+2)(s+3)}$

$$\frac{2s+1}{(s+1)(s+2)(s+3)} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+3}$$

$$\frac{2s+1}{(s+1)(s+2)(s+3)} = \frac{A(s+2)(s+3) + B(s+1)(s+3) + C(s+1)(s+2)}{(s+1)(s+2)(s+3)}$$

No.

Date:

$A(s+2)(s+3) + B(s+1)(s+3) + C(s+1)(s+2) = 2s+1$

$s=-1 \rightarrow A(-1+2)(-1+3) = 2(-1)+1$

$A(2) = -1$

$A = -\frac{1}{2}$

$s=-2 \rightarrow B(-2+1)(-2+3) = 2(-2)+1$

$B(-1) = -3.$

$B = 3$

$s=-3 \rightarrow C(-3+1)(-3+2) = 2(-3)+1$

$C(2) = -5$

$C = -\frac{5}{2}$

$f(t) = L^{-1}\left\{ \frac{-\frac{1}{2}}{s+1} + \frac{3}{s+2} - \frac{\frac{5}{2}}{s+3} \right\}$

$= -\frac{1}{2}L^{-1}\left[\frac{1}{s+1}\right] + 3L^{-1}\left[\frac{1}{s+2}\right] - \frac{5}{2}L^{-1}\left[\frac{1}{s+3}\right]$

$= -\frac{1}{2}e^{-t} + 3e^{-2t} - \frac{5}{2}e^{-3t}$

4. $F(s) = \frac{1}{s^2(s-3)}$

$\frac{1}{s^2(s-3)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-3}$

$1 = As(s-3) + B(s-3) + C(s^2)$

$s=0 \rightarrow B(0-3) = 1$

$B = -\frac{1}{3}$

$s=3 \rightarrow C(3^2) = 1$

$9C = 1$

$C = \frac{1}{9}$

$s=1 \rightarrow A(1)(1-3) + \left(-\frac{1}{3}\right)(1-3) + \frac{1}{9}(1)^2 = 1$

$-2A + \frac{2}{3} + \frac{1}{9} = 1$

$-2A = 1 - \frac{7}{9}$

$A = \frac{2}{9} \times -\frac{1}{2} = -\frac{1}{9}$

$f(t) = L^{-1}\left\{ \frac{-\frac{1}{9}}{s} - \frac{\frac{1}{3}}{s^2} + \frac{\frac{1}{9}}{s-3} \right\}$

$= -\frac{1}{9}L^{-1}\left[\frac{1}{s}\right] - \frac{1}{3}L^{-1}\left[\frac{1}{s^2}\right] + \frac{1}{9}L^{-1}\left[\frac{1}{s-3}\right]$

$= -\frac{1}{9} - \frac{1}{3}t + \frac{1}{9}e^{3t}$

No.

Date:

5. $F(s) = \dfrac{s}{(s-1)(s^2+1)}$

$$\frac{s}{(s-1)(s^2+1)} = \frac{A}{s-1} + \frac{Bs+C}{s^2+1}$$

$$s = A(s^2+1) + (Bs+C)(s-1)$$

$s=1 \rightarrow \quad 1 = A(1^2+1)$

$$A = \frac{1}{2}$$

$s=0 \rightarrow \quad \frac{1}{2}(0^2+1) + C(0-1) = 0$

$$\frac{1}{2} - C = 0$$

$$C = \frac{1}{2}$$

$s=2 \rightarrow \quad \frac{1}{2}(4+1) + (2B+\frac{1}{2})(2-1) = 2$

$$\frac{5}{2} + (2B+\frac{1}{2}) = 2$$

$$2B + \frac{6}{2} = 1$$

$$B = \frac{2-3}{2} = -1/2$$

$$f(t) = L^{-1}\left\{\frac{\frac{1}{2}}{s-1} + \frac{(-\frac{1}{2}s) + \frac{1}{2}}{s^2+1}\right\}$$

$$= \frac{1}{2}L^{-1}\left\{\frac{1}{s-1}\right\} - \frac{1}{2}L^{-1}\left\{\frac{s}{s^2+1} - \frac{1}{s^2+1}\right\}$$

$$= \frac{1}{2}e^t - \frac{1}{2}(\cos t - \sin t)$$